ACME METALS

Digital Material Passport

ID **DMP-METAL-007** - *Version* **1.0.0**
Issue Date **2025-05-20** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	**Bridge Constructors Europe Ltd.** Engineering Plaza 456 75001 Paris FR procurement@bridgeconstructors.example.com	**Order** PO-98765 · **20000 kg** Pos. 5 · 2025-04-20 **Delivery** DN-12345 · **20000 kg** Pos. 1 · 2025-05-19

Certificate Receiver
Civil Infrastructure Authority
Government Building 789
Administration Wing
1000 Brussels
BE
inspections@infra-authority.example.gov.be

Product

Product Name	Structural Steel S420N Plate	*Name (EN)*	1.8902
Batch ID	H-45678-01	*Shape*	Plate - Width 2500 mm - Thickness 25 mm - Length 12000 mm
Surface Condition	Shot blasted and primed	*Product Norm*	EN 10025-3 (2019)
Weight	20000 kg	*Product Norm*	EN 1090-2 (2018)
Production Date	2025-05-18		
Country of Origin	DE		

Chemical Analysis

Heat Number **H-45678** - *Melting Process* **EAF+LF+VD** - *Casting Method* **ContinuousCasting** - *Casting Date* **2025-05-17** - *Sample Location* **Ladle**

Symbol	C	Mn	Si	P	S	Nb	V	Ti	CEV
Unit	%	%	%	%	%	%	%	%	%
Min		1							
Max	0.22	1.7	0.6	0.025	0.02	0.05	0.05	0.03	0.48
Actual	0.16	1.38	0.32	0.016	0.008	0.022	0.034	0.009	0.4

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.4%

Mechanical Properties

Tensile Strength[1]	*Symbol*	1	2	3	*Average*	*Min*	*Max*	*Spec Min*	*Spec Max*	*Method*	*Status*
Value [MPa]	Rm	548	550	552	550	548	552	520	680	EN ISO - 6892-1	✓

Yield Strength[1]	*Symbol*	1	2	3	*Average*	*Min*	*Max*	*Spec Min*	*Method*	*Status*
Value [MPa]	ReH	438	440	442	440	438	442	420	EN ISO - 6892-1	✓

Elongation after - fracture 3 specimens tested, gauge length 5.65#S_0, transverse to rolling direction	*Symbol*	1	2	3	*Average*	*Min*	*Max*	*Spec Min*	*Method*	*Status*
Value [%]	A	20.5	21	21.5	21	20.5	21.5	19	EN ISO - 6892-1	✓

Charpy V-notch - Impact Energy 3 specimens tested at -20°C, transverse to rolling direction	*Symbol*	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*	*Method*	*Status*
Value [J]	KV	63	65	67	65	63	67	2	40	EN ISO - 148-1	✓

[1] 3 specimens tested at 20°C, transverse to rolling direction

Supplementary Tests

Test	*Result / limits*	*Method*	*Status*
Ultrasonic Testing Class S2E2	Yes - No recordable indications exceeding acceptance criteria	EN 10160	✓
Through-thickness Properties	Z25	EN 10164	✓

Weldability	Yes - Satisfactory welding properties	Internal Method based on EN ISO - 15614-1	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10025-3:2019 and EN 10204:2004 type 3.1. The product complies with the Construction Products Regulation (EU) No 305/2011 and is suitable for use in structural applications according to EN 1090-2:2018, up to and including Execution Class EXC3.

Validated by **John Smith**, Quality Manager, Quality Assurance - 2025-05-20

Validated by **Maria Schmidt**, Quality Manager, Quality Assurance - 2025-05-20